

# ■# PAPER266: HUDF Primordial IGM Magnetic Field & UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{crit} = 10^{11}$ T

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**Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

■# PAPER266: HUDF Primordial IGM Magnetic Field — UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at  $B_{crit} = 10^{11}$  T <!-- UQFF calibration:  $\kappa = 5.0e-4 \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ,  $\beta_i = 6.1e-1$  -->

**Author:** Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** HUDFGalaxies.cpp → HUDFCriticalMagneticTerm (Session 72g, UQFF 2.0 Upgrade) **Date:** March 2026 **Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

## Abstract

The HUDFGalaxies MUGE equation contains the magnetic suppression factor  $\text{corr}_B = 1 - B/B_{crit}$  with  $B = 10^{-10}$  T (primordial intergalactic medium) and  $B_{crit} = 10^{11}$  T as defined in the original C++ module header. This factor mirrors the behaviour of a Type II superconductor entering its upper critical field  $H_{c2}$ : at  $B \rightarrow B_{crit}$ , the UQFF gravitational field is expelled from the system, just as magnetic flux is expelled from a superconductor at  $H_{c2}$ .

The **uniquely rare discovery** of this paper is the identification of  $B_{crit} = 10^{11}$  T as the **UQFF Gravitational Meissner Boundary** — above this threshold, the  $\text{corr}_B$  factor vanishes and UQFF gravity is completely quenched. The HUDF's ultra-weak primordial field  $B = 10^{-10}$  T places it at  $\text{corr}_B = 1 - 10^{-21} \approx 1$  (completely unquenched, fully-active UQFF). This represents the maximum possible UQFF gravitational activity — the HUDF is a cosmological benchmark for the **unquenched UQFF limit**. In contrast, neutron stars with surface fields  $B \sim 10^{11}$  T (e.g., Cas A, PAPER257) sit exactly at this critical boundary, explaining their anomalous UQFF behavior near the Force Equivalence Class transition.

## 1. System Parameters

| Parameter              | Symbol      | Value                | Units | Regime                    |
|------------------------|-------------|----------------------|-------|---------------------------|
| HUDF primordial IGM    | B_HUDF      | $10^{-10}$           | T     | Fully unquenched          |
| Neutron star surface   | B_NS        | $10^{10} - 10^{12}$  | T     | Approaching/at boundary   |
| Magnetar               | B_mag       | $10^{13} - 10^{15}$  | T     | Above boundary (quenched) |
| UQFF critical          | B_crit      | $10^{11}$            | T     | Meissner boundary         |
| QED Schwinger critical | B_Schwinger | $4.4 \times 10^{13}$ | T     | QED pair-production       |

## 2. Gravitational Meissner Effect

## 2.1 The Magnetic Suppression Factor

The  $\text{corr}B = (1 - B/B_{\text{crit}})$  term controls the amplitude of UQFF gravitational acceleration:

$$U_{\{g4\}} = U_{\{g1\}} \cdot (1 - B/B_{\text{crit}}) = U_{\{g1\}} \cdot \text{corr}_B$$

The UQFF total:

$$U_{\{g, \text{UQFF}\}} = (U_{\{g1\}} + U_{\{g4\}}) \cdot (\dots) = U_{\{g1\}} \cdot \left(2 - \frac{B}{B_{\text{crit}}}\right) \cdot (\dots)$$

## 2.2 Phase Diagram by $B/B_{\text{crit}}$

| System                     | B (T)                  | B/B <sub>crit</sub> | corr <sub>B</sub> | UQFF regime          |
|----------------------------|------------------------|---------------------|-------------------|----------------------|
| HUDF primordial IGM        | 10 <sup>-10</sup>      | 10 <sup>-21</sup>   | ≈ 1.0             | <b>Fully active</b>  |
| Galaxy cluster (Faraday)   | 10 <sup>12</sup>       | 10 <sup>-1</sup>    | ≈ 1.0             | Fully active         |
| Solar wind near Earth      | 5×10 <sup>-10</sup>    | 5×10 <sup>-20</sup> | ≈ 1.0             | Fully active         |
| Neutron star (Cas A)       | 10 <sup>12</sup>       | 10 <sup>-3</sup>    | 0.999             | Nearly active        |
| PSRJ0030 pulsar            | 3×10 <sup>12</sup>     | 3×10 <sup>-3</sup>  | 0.997             | Nearly active        |
| B <sub>crit</sub> boundary | <b>10<sup>11</sup></b> | <b>1.0</b>          | <b>0.0</b>        | <b>QUENCH POINT</b>  |
| Strong magnetar            | 10 <sup>13</sup>       | 100                 | -99               | Negative: unphysical |

**Note:**  $B > B_{\text{crit}}$  in C++ original has  $\text{corr}B < 0$  (unphysical). The validator in `HUDFCriticalMagneticTerm` permits  $B \leq B_{\text{crit}} \times 1.1$  for probing the near-critical zone without full sign reversal.

## 2.3 Meissner Analogy

In Type II superconductors, the order parameter  $\psi$  (Cooper pair density) follows:

$$|\psi|^2 \propto \left(1 - \frac{B}{B_{c2}}\right) \quad \text{near } H_{c2}$$

The UQFF factor  $\text{corr}B = (1 - B/B_{\text{crit}})$  is **structurally identical** to this mean-field suppression, with  $\text{corr}_B$  playing the role of  $|\psi|^2$ . This suggests:

$$U_{\{g, \text{UQFF}\}} \propto |\psi|^2_{\text{UQFF}} = 1 - B/B_{\text{crit}}$$

The UQFF gravitational quantum field is a condensate analogous to a superconducting order parameter. As  $B \rightarrow B_{\text{crit}}$ , the condensate melts — gravity quenches.

## 2.4 Critical Field Derivation

$B_{\text{crit}} = 10^{11}$  T corresponds to:

**Neutron star polar cap:** Standard pulsar surface field  $B_s \sim 10^{12} - 10^{13}$  T; at the critical boundary  $B_s = B_{\text{crit}} = 10^{11}$  T, UQFF gravity is 99.9% active ( $\text{corr}B \approx 1 - B_s/B_{\text{crit}} \approx 0$  for  $B_s = 10^{11}$  T)

**Landau level spacing:**  $\hbar\omega_c = \hbar(eB/m_e c) = e\hbar B/m_e c$ ; at  $B = 10^{11}$  T  $\rightarrow \hbar\omega_c \approx 1.15 \times 10^{-2}$  J  $\approx 72$  MeV — near pion mass scale, suggesting  $B_{\text{crit}}$  marks the hadronic confinement–deconfinement boundary in QCD

**UQFF LENR coupling:** At  $B = B_{\text{crit}}$ , the 1.25 THz LENR oscillator resonance condition is modified by cyclotron resonance  $\omega_{\text{LENR}} = \omega_c(B_{\text{crit}})$  for  $e^-$  at  $B_{\text{crit}} \approx 1.25 \times 10^{12}$  rad/s  $\rightarrow B \approx 7 \times 10^{-3}$  T. The mismatch confirms  $B_{\text{crit}} = 10^{11}$  T is a separate, purely UQFF-gravitational boundary.

### 3. Gravitational Meissner Effect Theorem

**Theorem (UQFF Gravitational Meissner Effect):** The UQFF gravitational field  $\mathcal{G}(B)$  obeys a Meissner-like suppression law:

$$\mathcal{G}(B) = \mathcal{G}_0 \cdot \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

where  $\mathcal{G}_0 = U_{g1}(2 - B/B_{\text{crit}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$  evaluated at  $B = 0$ .

The field  $\mathcal{G}$  is **expelled** from the UQFF medium at  $B = B_{\text{crit}}$  (*gravitational quench*), analogous to magnetic flux expulsion from a superconductor at  $H_{c2}$ .

**Corollary 1 (HUDEF Maximum):** The HUDEF with  $B_{\text{HUDEF}} = 10^{-10} \text{ T}$  gives  $\text{corr}B = 1 - 10^{-21} \approx 1$ , representing the **maximum UQFF gravitational activity achievable** in a cosmic environment. This makes the HUDEF the benchmark calibration field for fully-active UQFF.

**Corollary 2 (NS Critical Zone):** Neutron stars with  $B \sim 10^{11} \text{ T}$  sit at the Meissner boundary. PSRJ0030 and Cas A (PAPER255, 257) with  $B \sim 10^{10} - 10^{11} \text{ T}$  are within 2–3 orders of  $B_{\text{crit}}$  — within the "upper critical zone" where UQFF is suppressed by 0.1–1%, explaining the slight deviation of their  $F_{\text{UBi}_i}$  from the fully unquenched theoretical maximum.

**Corollary 3 (Magnetar Above-Critical):** Magnetars with  $B > B_{\text{crit}}$  would have  $\text{corr}B < 0$  — a physically distinct phase where  $U_{g4}$  contributes positively to gravity reversal, potentially explaining the anomalous braking indices of highly magnetised NSs.

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### 4 Observational Predictions

**HUDEF  $z = 3.5$  ( $B \approx 10^{-10} \text{ T}$ ):**  $\text{corr}B \approx 1.0 \rightarrow \text{maximum } F_{\text{UBi}_i}$ . ALMA Faraday rotation measurements of HUDEF background sources at  $z > 3$  can constrain  $B_{\text{HUDEF}}$  and verify  $\text{corr}B \approx 1$ .

**Cas A neutron star ( $B \approx 10^{11} \text{ T}$  thermal-scale):** From PAPER257,  $B = 10^{11} \text{ T} \rightarrow \text{corr}B = 1 - 10^{-1} \approx 1$ . Even with this tiny surface field in the PAPER\_257 model, Cas A remains fully unquenched.

**Magnetar quench test:** An X-ray polarimetry observation of a magnetar with  $B > 10^{11} \text{ T}$  (e.g., SGR 1806-20,  $B \sim 2 \times 10^{11} \text{ T}$ ) should show suppressed UQFF signature compared to the HUDEF benchmark — a direct test of the Gravitational Meissner Effect.

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### 5. References

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